DSA2102 Numerical Computation

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2026

Chapter 3 | Least Square Problems

Introduction

Suppose that you have a system $Ax = b$, where A is a $n \times m$ matrix where $n > m$. In such a system, we usually say that the system is overdetermined, and a solution wouldn't exist. Often times when presented with a problem like this, such as fitting a line over a couple of data points (elaborated further in later chapters concering interpolation,) we would like to find some solution that best fit our data.

Let's take a look at our previous example $Ax = b$. Suppose for $\mathbb{R}^2$we write them out as such:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

Here, we couldn't find an exact solution to the problem. But, we can come up with some solution that minimize the 'error'. Such a problem is called a least square problem.

To understand how we could solve such a problem, we'll analyze $Ax = b$ once more. If we think of A as as some plane in $\mathbb{R}^2$, we can think of the existance of a solution for $Ax = b$ as that b lies somewhere within the span of A. Likewise, if there is no solution, b lise somewhere outside the span of A. Notice, however, that we can find some vector y within the span of A ($A\hat{x} = y$) such that $b = y + e$ for some vector e taht satisfies $A^T e = 0$ (e is perpendicular to the plane of A). As such, we can write the following:

$$0 = A^T e = A^T(b - y) = A^T(b - A\hat{x}).$$

$$A^T b = A^T A\hat{x}.$$

We can interpret the solution as finding the vector $\hat{x}$ that minimizes e (as it so happens for $\mathbb{R}^2$), the minimum is just the projection of b unto the plane.

QR Factorization

A QR factorization is simply decomposing a vector into $A = QR$, where Q is a orthogonal matrix $Q = Q^T$ and R is an upper triangular matrix. We could interpret this as taking an orthonormal basis Q and constructing A using R. Generally, it is preffered over LU as it is more stable and does not require pivoting as LU does.

When solving a QR factorization, we have a couple of options:

1. Gramm-Schmidtt Algorithm
2. Householder Reflections
3. Givens Rotation
4. Singular Value Decomposition (SVD)

Overall, Gramm-Schmidtt is not preffered as it quite slow. We'll talk about the first three in this chapter, and SVD in the later chapters concerning eigenvalue problems.

Gramm-Schmidtt

The idea behind Gramm-Schmidtt is to construct a set of orthonormal vectors by considering each columns of the matrix A, and iteratively subtracting it by it's projections on each previous columns and normalizing it, while storing the dot product of each vectors in an upper triangular matrix.

As one might expect, when doing the calculation by relying on previous results of computing each row of Q, error would be magnified. As such, when calculating a QR decomposition using this method, we would design the algorithm to iterate by rows instead.

Rotations and Reflections

Chapter 4 | Eigenvalue Problems

Recap

To recap, for a square matrix A and a non-zero vector x:

$$\begin{aligned} Ax &= \lambda x \\ 0 &= (A - \lambda I)x \\ p(\lambda) &= \det(A - \lambda I) = 0. \end{aligned}$$

The solution to x is called the Eigenvectors, and solution to λ are the Eigenvalues. $p(\lambda)$ is called the characteristic polynomial. $A - \lambda I$ is singular. This leads to the spectral theorem for symetric real matrices, which states that

Spectral Theorem for Real Matrices

A Symetric real matrix A can be diagonalized using an orthonormal basis of vectors.

$$A = QDQ^T.$$

A is said to be orthogonally diagonalizable.

This means that for any symetric real matrix, there always exists an orthonormal basis (Q) such that A is a simple diagonal matrix. Intuitively, this is changing the basis for A, such that A becomes simple scaling.

Polynomials and Eigenvalues

In numerical computation, the bulk of finding the root of a polynomial can be computed by computing the Eigenvalues, and such problem is classified as an eigenvalue problem. We do this by computing the eigenvalues using teh companion matrix, which is just the matrix form of the (monic, ie. the coeffecient for the largest term is 1) characteristic polynomial, given by

$$C_n = \begin{pmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{n-1} \end{pmatrix}.$$

For small polynomial ($n \leq 4$), we can just use the readily available formula to compute the root. The Abel-Ruffini Theorem states that there is no general formula for finding the root of polynomial of power ≥ 5 (TODO: Read), which is very common, and we would need to rely on algorithm to solve it iteratively, and thus find the root of the polynomial indirectly. Generally, such algorithm solves the root one at a time, so calculating all the root at once usually boils down to solving the eigenvalues.

Before doing so, let's take a look at the question of conditioning, uniqueness and existence of eigenvectors and eigenvalues in the real field. In general,

1. For an upper triangular matrix, the diagonal entries is the eigenvalues.
2. All symetric matrices have atleast one eigenvector and eigenvalues.
3. Every scalar multiple of an eigenvector is an eigenvector.
4. Every eigenvalues of a matrix is unique.
5. The algebraic multiplicity of a matrix A is $det(A - \lambda I)$.
6. The geometric multiplicity $dim(A - \lambda I)$ is equal to the number of unique eigenvalues of A. It is equal or less than the algebraic multiplicity. If it is strictly less, the matrix A is defective. Non-defective matrices of size $n \times n$ (ie. those with n linearly independent eigenvectors) are diagonalizable.

$$AX = XD \iff X^{-1}AX = D \iff A = XDX^{-1},$$

where X is a matrix of eigenvectors, and D is a diagonal matrix of the eigenvalues. If all the collumns in X is orthogonal, then we say that A is orthonormaly diagonalizable, and by the spectral theorem, $A = A^T$. More often than not, such matrices are symetric.

Operations on a matrix changes the eigenvalues. In general, the following applies:

1. Shifts $Ax = \lambda x \to (A - \sigma I)x = (\lambda - \sigma)x$
2. Inverses $Ax = \lambda x \to A^{-1}x = \frac{1}{\lambda}x$
3. Powers $Ax = \lambda x \to A^k x = \lambda^k x$
4. Polynomials $Ax = \lambda x \to P(A)x = P(\lambda)x$
5. Cholseky, QR, Elemination does nto preserve the non-zero eigenvalues.

The condition number (2-norm) for a non-defective eigenvalue problem is $X^{-1}AX = D$ is given by the condition numebr (2-norm) of X.

Singular Value Decomposition

Singular Value Decomposition

For $A \in \mathbb{R}^{m \times n}$,

$$A = U\Sigma V^T,$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal. The matrices $A^T A$ and AA^T are real symetric and orthogonally diagonalizable. Σ is a diagonal matrix consisting of the singular value of A (the square root of AA^T and $A^T A$ non-zero eigenvalues.)

When decomposing a matrix with SVD, we typically try to solve either $A^T A$ or AA^T for their eigenvalues and eigenvectors. Typically, we choose whichever is least complex to compute.

After calculating the eigenvalues and eigenvector, and forming either U or V, we would need to form the other base.

Power Iteration and Variants

The idea behind power iteration is quite simple: For a matrix A that we would like to compute the eigenvalues and eigenvectors, we start with a vector v_0 (random, most like,) and repeatedly applying Av_i iteratively.

If A is non-defective and the largest eigenvalue is strictly larger than the other ones, we can analyze the algorithm's convergence quite easily:

Tutorial 6

Question 1

Use the Intermediate Value theorem to find an interval of length one that contains a root of the equation,

(a) $x^3 = 9$

(b) $\ln x + x^2 = 3$

Guess an root exists at [2,3]

$$f(2) = 8 - 9 = -1$$
$$f(3) = 27 - 9 = 18$$
$$.$$

By IVT, a root exists within that interval. Do the same thing for the second part: We guess an interval of [1,2]

$$f(1) = \ln 1 + 1 - 3 = -2$$
$$f(2) = \ln 2 + 4 - 3 > 0.$$

By IVT, a root exists within that interval.

Question 2

Consider the function $f(x) = x^3 + x - 1$

1. Write down the formula for x_k in terms of x_{k-1} for the Newton iteration.
2. Starting with an initial guess of $x = -0.7$, perform two iteration of the Newton method

Question 3

Find the eigenvalue of the matrix

$$\begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}.$$

$$\det \begin{pmatrix} 1-\lambda & 1 \\ 2 & 4-\lambda \end{pmatrix} = 0$$
$$(1-\lambda)(4-\lambda) + 2 = 0$$
$$(\lambda - 3)(\lambda - 2) = 0.$$
$$\lambda \in \{2, 3\}.$$

Question 4

Give an example to show that the eigenvalues can be changed when a multiple of one row is subtracted from another. Why is a zero eigenvalue not changed by the steps of elimination?

Take the matrix A from Q4. Suppose that we perform $R_2 \to R_2 - 2R_1$ on the matrix. We would have

$$\begin{pmatrix} 1 & -1 \\ 0 & 6 \end{pmatrix}.$$

Should we calculate the eigenvalue, we would get $\lambda \in \{1, 6\}$, clearly different than the original result.

To explain the reasoning behind the zero eigenvalues not being affected, we'll go back to the definition of eigenvalues and eigenvectors. For a matrix A with an eigenvalue $\lambda = 0$, we have

$$\det(A - \lambda I) = 0.$$

For any $\lambda \neq 0$, an elimination would shift the determinant of A such that the value of λ would need to compensate. For a $\lambda = 0$ however, $\det(A)$ is already 0 (singular), and so any row operations would not change the determinants, and λ need not compensate.

Question 5

Compute $A^T A$ and AA^T and their eigenvalues and unit eigenvectors for

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Use your result to factor $A = U\Sigma V^T$

For $A^T A$, we have

$$A^T A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\det \begin{pmatrix} 1-\lambda & 1 & 0 \\ 1 & 2-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{pmatrix} = (1-\lambda)^2(2-\lambda) - 2(1-\lambda) = 0.$$

$$\lambda \in \{0, 1, 3\}.$$

Calculating the unit eigenvectors, we have

$$v_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} v_3 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

$$V = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

Calculating U,

$$v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} v_3 = \frac{1}{\sqrt{3}} \frac{1}{\sqrt{6}} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\frac{1}{\sqrt{2}}} \end{pmatrix}.$$

Putting it all together, we have

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\frac{1}{\sqrt{2}}} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

Question 6

Let A be a $m \times n$ matrix with real entries and consider teh symetric eigenvalue problem

$$\begin{pmatrix} 0 & A \\ A^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \lambda \begin{pmatrix} u \\ v \end{pmatrix}.$$

Show that if λ, u, v satisfy the relationship above and $|u| = 1 = \|v\|$, then $\|\lambda\|$ is a singular value of A with corresponding left and right singular vectors u and v, respectively.

Breakdown teh expression:

$$A^T u = \lambda v, \quad Av = \lambda u.$$

Then,

$$AA^T u = \lambda^2 u \quad A^T A v = \lambda^2 v,$$

thus showing that λ is a singular value of A.

Question 7

The Fibonnaci Sequence $F_1, F_2, \ldots$is defined by $F_1 = 1, F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. Define A by $A(x, y)^T = (y, x + y)^T$

(a) Show that $A^n(0, 1)^T = (F_n, F_{n+1})^T$ for each positive integer n.

(b) Find the eigenvalues of A.

(c) Find the basis of $\mathbb{R}^2$ consisting of eigenvectors of A.

(d) Use part (c) to compute $A^n(0, 1)^T$. Conclude that

$$F_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right).$$

(e) Use part (d) to conclude that for each positivei nteger n, F_n is the integer closest to

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n.$$

Starting with the first part. We'll use induction to prove it such. The base case is provided by $n = 1$, and

$$A\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}F_1\\F_2\end{pmatrix}.$$

For the inductive steps, let $n = m + 1$. Asumming that the proposition holds for $n = m$

$$\begin{aligned} A^{m+1}\begin{pmatrix}0\\1\end{pmatrix} &= A\begin{pmatrix}F_m\\F_{m+1}\end{pmatrix} \\ &= \begin{pmatrix}F_{m+1}\\F_m + F_{m+1}\end{pmatrix} \\ &= \begin{pmatrix}F_{m+1}\\F_{m+2}\end{pmatrix}. \end{aligned}$$

so the proposition holds. For the eigenvalues of A, suppose that

$$A\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}\lambda_0 & 0\\0 & \lambda_1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$

Then,

$$\begin{pmatrix}y\\x+y\end{pmatrix} = \begin{pmatrix}\lambda_0 & 0\\0 & \lambda_1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$

$$\lambda^2 x - \lambda x - x = 0 \implies \lambda^2 - \lambda - 1 = 0.$$

Solving the equation would yield the eigenvalues. We find

$$\lambda = \frac{1 \pm \sqrt{5}}{2}.$$

We can then proceed to find the eigenvectors of A. From our previous calculation, we find

$$\begin{pmatrix}y\\x+y\end{pmatrix} = \begin{pmatrix}\lambda_0 & 0\\0 & \lambda_1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$

Chapter 5: Interpolation

We'll start out with polynomial interpolation, which is basically try to fit a polynomial such that it matches the given data points (we'l skip over trig and rational interpolations in this course.)

In general, for a polynomial of degree d, a solution exists when the number of the given data points is no more than $d+1$, and the solution is unique as long as there is exactly $d+1$ points. These all depends on the basis matrix. One can choose any basis, each with their own advantages and disadvatages. I'll start with the monomial basis matrix, also known as the Vandermonde matrix. It looks something like this:

$$X = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^d \\ 1 & x_1 & x_1^2 & \cdots & x_1^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{pmatrix} = \begin{pmatrix} M_0(x_0) & M_1(x_0) & M_2(x_0) & \cdots & M_d(x_0) \\ M_0(x_1) & M_1(x_1) & M_2(x_1) & \cdots & M_d(x_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ M_0(x_n) & M_1(x_n) & M_2(x_n) & \cdots & M_d(x_n) \end{pmatrix}.$$

This looks like

$$\begin{aligned} p(x) &= a_0 + a_1 x^2 + \ldots + a_n x^d \\ &= X\overline{a}. \end{aligned}$$

Example Find the polynomial that fit the points $(-2,-27),(0,-1),(1,0)$.

$$\begin{pmatrix} 1 & x_0 & x_0^2 \\ 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & -2 & 4 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -27 \\ -1 \\ 0 \end{pmatrix}.$$

Let's consider another basis: the Langrange. S

Consider another basis: the Newton.